

TASK 5:

Ticket class:

```
app > ticket.py > Ticket
1 from dataclasses import dataclass
2 from datetime import date
3 import sqlite3
4
5 You, 43 minutes ago | 1 author (You)
6 @dataclass
7 class Ticket:
8     ticket_id: str
9     student_number: str
10    student_name: str
11    category: str
12    description: str
13    priority: str
14    status: str = "Open"
15    date_logged: str = str(date.today())
16    closing_comment: str = ""
17
18 def generate_ticket_id():
19     conn = sqlite3.connect("data/tickets.db")
20     cursor = conn.cursor()
21     cursor.execute("SELECT ticket_id FROM tickets ORDER BY ticket_id DESC LIMIT 1")
22     last = cursor.fetchone()
23     conn.close()
24
25     if last:
26         num = int(last[0][1:]) + 1
27         return f"T{num:03d}"
28     else:
29         return "T001"
```

Sample database:

The screenshot shows a code editor with several tabs at the top: 'database.py M X', 'tickets.db U', 'ticket.py M', 'main.py M', and '__init__.py U'. The active tab is 'database.py', and the cursor is at line 26. The code defines a function 'init_db()' that connects to a SQLite database 'data/tickets.db', creates a table 'tickets' with columns 'ticket_id', 'student_number', 'student_name', 'category', 'description', 'priority', 'status', 'date_logged', and 'closing_comment', and then commits and closes the connection. A main block calls 'init_db()' and prints a success message.

```
1 import sqlite3
2
3 def init_db():
4     conn = sqlite3.connect("data/tickets.db")
5     cursor = conn.cursor()
6
7     cursor.execute("""
8         CREATE TABLE IF NOT EXISTS tickets (
9             ticket_id TEXT PRIMARY KEY,
10            student_number TEXT NOT NULL,
11            student_name TEXT NOT NULL,
12            category TEXT NOT NULL,
13            | description TEXT NOT NULL,
14            | priority TEXT NOT NULL,
15            | status TEXT NOT NULL,
16            date_logged TEXT NOT NULL,
17            closing_comment TEXT
18        );
19    """)
20
21    conn.commit()
22    conn.close()
23
24 if __name__ == "__main__":
25     init_db()
26     print("Database initialized successfully.")
```

This structure is suitable for the tickets and all necessary data types are chosen for the table for example student ID must start with ST so I made it text and the same with ticket ID.